From Three-Layer Unlearning Impact to Forget-Set Audit: Predicting Impact Profiles from Forget-Set Geometry Alone

Anonymous EMNLP submission

Abstract

Large language model (LLM) unlearning must remove target knowledge represented by a forget set $\mathcal{D}_f$ while limiting collateral damage to useful knowledge. Existing evaluations usually collapse this requirement into aggregate forget/retain scores or compare algorithms on a fixed forget split, leaving unclear how unlearning impact spreads from the target set to neighbouring and unrelated text, and whether damaging forget sets can be identified before running unlearning. We construct 100 WikiText-103 triplets, unlearn Llama-3.1-8B-Instruct with GradAscent for each forget set, and evaluate all checkpoints against all triplets in a 100×100 cross-PPL matrix, yielding 1.5M per-sample perplexity ratios. We represent each run by a *three-layer impact profile*: intended degradation on $\mathcal{D}_f$ (L_F), neighbour impact on held-out same-topic samples (L_N), and bystander impact on cross-topic samples (L_B); we then predict this profile from a 12-feature intrinsic geometry descriptor of $\mathcal{D}_f$ using a leave-one-out Random Forest auditor. Impact decays across layers (L_F=1.96×, L_N=1.32×, L_B=1.19×), but spillover does not vanish: 73.8% of cross-topic samples have elevated PPL, and L_F varies by 1.59× across forget sets under the same algorithm and hyperparameters. The geometry-only auditor gives rank-significant predictions for all layers (Spearman ρ=0.56/0.87/0.59 for $L_F/L_N/L_B$), and an independent NPO replication remains rank-significant (ρ=0.66/0.82/0.51, all CIs above zero). Unlearning side effects are partly a property of the forget set itself, and forget-set geometry can provide a cheap pre-screen for risky candidates before full unlearning evaluation.

1 Introduction

After deployment, large language models are increasingly asked to remove specific knowledge. Copyright disputes, privacy regulation, user deletion requests, and the discovery of toxic, private, or copyrighted content in pretraining corpora all create cases where a model should stop reproducing or relying on a target forget set $\mathcal{D}_f$ [Maini et al., 2024, Shi et al., 2024, Dorna et al., 2025]. Since retraining from scratch is usually impractical, *LLM unlearning* aims to suppress this target knowledge through post-training updates while preserving the rest of the model. This preservation requirement is critical: in deployed systems, a successful deletion is not sufficient if the same update silently corrupts nearby facts, paraphrases, or unrelated capabilities.

Existing unlearning evaluations do not yet expose this side effect with enough resolution. Most studies report aggregate utility or a small set of post-hoc probes, which can miss the *geometry* of unlearning damage: how impact changes as evaluation text moves from the forget set itself, to same-topic neighbours, to different-topic bystanders. Recent knowledge-hole probing work shows that benchmark-level utility can remain high even when adjacent or latent knowledge is corrupted [Ko et al., 2025]. At the same time, most benchmarks fix the forget split and compare algorithms, treating the choice of $\mathcal{D}_f$ as given. This leaves a practical question unanswered: among many candidate forget sets, can we identify which ones are likely to cause broader damage before running expensive unlearning?

We address this question with a controlled cross-evaluation framework for three-layer unlearning impact. We cluster WikiText-103 passages into semantic groups, sample 100 candidate triplets, and split each triplet into disjoint train, validation, and test texts. For each triplet, the train split is the forget set $\mathcal{D}_f$, while validation and test texts provide held-out same-topic neighbours. After unlearning Llama-3.1-8B-Instruct with a fixed GradAscent configuration, we evaluate every unlearned checkpoint on every triplet, yielding a cross-PPL matrix with 1.5M per-sample perplexity ratios. This matrix separates **intended degradation** on the forget set (L_1), **locality damage** on same-topic held-out text (L_2), and **spillover damage** on unrelated-topic bystanders (L_3). The core insight is that unlearning impact is structured by semantic distance, but the chosen forget set also has an intrinsic geometry that helps predict how severe that impact will be.

Our measurements show both distance decay and persistent collateral damage. Across the cross-PPL matrix, impact decreases strictly across the three layers ($L_1 > L_2 > L_3$), yet it does not vanish: L_3 remains above the base model on 73.8% of cross-topic samples evaluated against unrelated clusters. Damage magnitude also varies substantially with the chosen forget set, even under the same algorithm and hyperparameters; L_1 varies by $1.59\times$ across the 100 candidates. We then formulate *forget-set audit*: predicting the three-layer impact profile from forget-set geometry alone. Using a 12-dimensional intrinsic geometry descriptor computed from the 50 forget-set embeddings, a Random Forest regressor obtains rank-significant predictions under leave-one-out cross-validation, with Spearman correlations of 0.56, 0.87, and 0.59 for L_1, L_2, and L_3; all 95% bootstrap confidence intervals exclude zero. The auditor runs in seconds and serves as a pre-screen for forget-set selection before full unlearning.

Our contributions are: (1) We introduce a measurement framework for three-layer unlearning impact, separating intended forget-set degradation, same-topic locality damage, and cross-topic spillover damage within a unified cross-PPL matrix. (2) We provide quantitative evidence that unlearning impact decays with semantic distance, yet collateral damage remains non-zero even far from the forget set: $L_3 > 1$ on 73.8% of cross-topic samples evaluated against unrelated clusters. (3) We formulate *forget-set audit*, which predicts the three-layer impact profile from forget-set geometry alone. A 12-feature Random Forest auditor achieves rank-significant predictions across all three layers, including $\rho = 0.87$ and $R^2 = 0.74$ for locality damage, enabling cheap screening before full unlearning.

Sec. 2 formalises the three-layer impact and audit problems. Sec. 3 describes the triplet schema, unlearning procedure, and cross-PPL matrix. Sec. 4 reports the three-layer impact findings. Sec. 5 presents the audit features, forget-set audit, and predictor. Sec. 6–7 discuss limitations and outlook. Sec. 8 closes with related work.

2 Problem Formulation

Experimental objects. Let M_{θ_0} denote the original language model before unlearning, and let $\mathcal{U}$ be a fixed post-training unlearning algorithm. We consider a collection of candidate semantic groups $\mathcal{G} = \{G_i\}_{i=1}^N$, where each group contains passages about a coherent topic. Each group is split into three disjoint subsets: G_i^{tr}, G_i^{val}, and G_i^{te}. For candidate i, the forget set and resulting unlearned checkpoint are

$$\mathcal{D}_f^{(i)} = G_i^{\text{tr}}, \tag{1}$$

$$M_{\theta_i} = \mathcal{U}(M_{\theta_0}, \mathcal{D}_f^{(i)}). \tag{2}$$

The experiment fixes M_{θ_0} and $\mathcal{U}$, varies only $\mathcal{D}_f^{(i)}$, and compares the resulting checkpoints across the same semantic groups.

Measurement metric. For an evaluation passage x, we measure unlearning impact by the base-normalised perplexity ratio

$$r_i(x) = \frac{\text{PPL}(M_{\theta_i}, x)}{\text{PPL}(M_{\theta_0}, x)}, \tag{3}$$

$$R_i(S) = \exp\left(\frac{1}{|S|}\sum_{x\in S} \log r_i(x)\right). \tag{4}$$

A value $r_i(x) > 1$ means that unlearning $\mathcal{D}_f^{(i)}$ makes the model assign lower likelihood to x than the original model did. For a set of passages S, $R_i(S)$ is the geometric mean ratio. Normalising by M_{θ_0} makes measurements comparable across passages with different base difficulty.

Three-layer impact profile. For each candidate forget set $\mathcal{D}_f^{(i)}$, we separate impact into three semantic layers. Let $\mathcal{N}^{(i)} = G_i^{\text{val}} \cup G_i^{\text{te}}$ be the held-out same-topic neighbour set, and let $\mathcal{B}^{(i)} = \bigcup_{j\neq i} G_j^{\text{te}}$ be the cross-topic bystander set. The three-layer impact profile is

$$L_1^{(i)} = R_i(G_i^{\text{tr}}) \qquad \text{(intended degradation)}, \tag{5}$$

$$L_2^{(i)} = R_i(\mathcal{N}^{(i)}) \qquad \text{(locality damage)}, \tag{6}$$

$$L_3^{(i)} = R_i(\mathcal{B}^{(i)}) \qquad \text{(spillover damage)}, \tag{7}$$

$$\mathbf{y}^{(i)} = \left(L_1^{(i)}, L_2^{(i)}, L_3^{(i)}\right). \tag{8}$$

High L_1 is the intended forgetting signal; high L_2 or L_3 indicates collateral damage. In later tables we use the mnemonic aliases L_F, L_N, L_B for these same three layers.

Forget-set audit. Measuring $\mathbf{y}^{(i)}$ requires running $\mathcal{U}$ and evaluating the resulting checkpoint. We formulate *forget-set audit* as a pre-unlearning prediction task: given only the candidate forget set $\mathcal{D}_f^{(i)}$, predict

its future three-layer impact profile. Let $\phi(\mathcal{D}_f^{(i)}) \in \mathbb{R}^d$ be an intrinsic descriptor computed from the forget-set passages. An auditor $A_\psi : \mathbb{R}^d \to \mathbb{R}^3$ is trained with the objective

$$\hat{\psi} = \arg\min_{\psi} \sum_{i=1}^{N} \left\| A_\psi\left(\phi(\mathcal{D}_f^{(i)})\right) - \mathbf{y}^{(i)} \right\|_2^2. \quad (9)$$

At prediction time, the auditor has no access to M_{θ_i} or to post-unlearning evaluation results. Its practical goal is to estimate or rank which candidate forget sets are likely to induce larger intended degradation, same-topic locality damage, or cross-topic spillover damage before paying for full unlearning and cross-evaluation.

3 Experiment Setup

3.1 Triplet Schema

We start from WikiText-103 [Merity et al., 2017], filter to 16,038 usable passages, and embed them with `all-MiniLM-L6-v2` [Wang et al., 2021] into 384-dim vectors. HDBSCAN [Campello et al., 2013] with `min_cluster_size`=200, `min_samples`=5, Euclidean metric, and EOM cluster selection produces 10 semantic clusters (covering recurring WikiText topics: Federer, Harry Potter, tennis matches, etc.) plus a noise pool of 9,198 (57.4%). Fig. 1 summarises the construction.

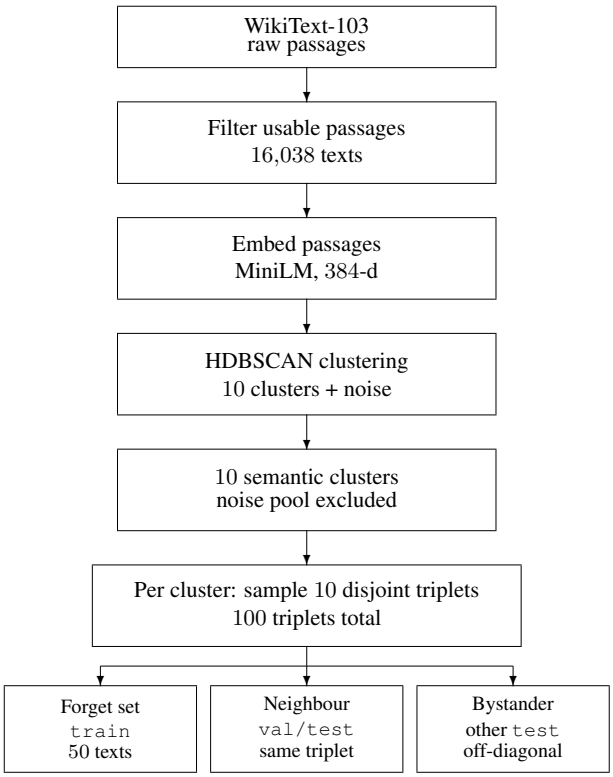

Figure 1: Triplet schema. WikiText-103 passages are filtered, embedded, and clustered; each semantic cluster yields 10 triplets. The train split is the forget set, validation/test splits are same-topic neighbours, and test splits from other triplets serve as cross-topic bystanders in off-diagonal evaluation.

For each cluster we sample 10 disjoint *triplets*, giving 100 triplets in total. Each triplet contains three same-cluster but mutually disjoint splits of 50 texts:

Split	50 texts	Downstream role
`train`	forget set $\mathcal{D}_f$	L_F: intended degradation
`validation`	neighbours	retain loss + L_N
`test`	probe texts	diag L_N; off-diag L_B

The within-cluster but mutually disjoint design lets us read off three semantic distances by simply re-indexing the same evaluation matrix. Different triplets may overlap across clusters (sampling is independent), but within a triplet the three splits never share a text.

3.2 Unlearning Procedure

We unlearn Llama-3.1-8B-Instruct [Dubey et al., 2024] with the GradAscent objective [Jang et al., 2023]: for each $\mathcal{D}_f$ we maximise the negative log-likelihood on $\mathcal{D}_f$ while keeping a small retain-loss term on the same-cluster validation split. Hyperparameters follow defaults from OpenUnlearning [Dorna et al., 2025]: learning rate $1{\times}10^{-5}$ with linear decay, paged AdamW (32-bit), per-device batch size 8 with gradient accumulation 8 (effective batch 64), 5 epochs (resulting in 5 optimiser steps because $|\mathcal{D}_f|{=}50 < 64$), warmup of 1 step, weight decay 0.01, BF16. All runs use a single H100 80 GB. Each unlearn run takes ≈ 75 s wall-clock; the full $n{=}100$ sweep finishes in 1 h 54 min and produces ≈ 1.5 TB of checkpoints (≈ 15 GB each). Train loss converges in $[-0.84, -0.62]$ (negative is expected for GradAscent).

3.3 Cross-PPL Matrix and Three Layers

Given 100 unlearned checkpoints and 100 evaluation triplets, we evaluate every checkpoint on every triplet's three splits, yielding $100 \times 100 \times 3 \times 50 = 1{,}500{,}000$ per-sample perplexities (Fig. **??**; numerically: 5,000 L_F samples on the diagonal train splits, 10,000 L_N samples on diagonal val/test, and 495,000 L_B samples on the off-diagonal test). We pair each measurement with the corresponding base-model PPL and define

$$r_{m,e,s,i} = \frac{\mathrm{PPL}_{\text{unlearned},m}(x_i^{e,s})}{\mathrm{PPL}_{\text{base}}(x_i^{e,s})}, \quad (10)$$

	L_F	L_N	L_B
Samples	5,000	10,000	495,000
Geo-mean r	1.96×	1.32×	1.19×
% samples with r>1.1	100.0	90.8	73.8
% samples with r>2.0	42.0	2.4	0.8

Table 1: Three-layer unlearning impact headline (n=100 forget sets, 1.5M per-sample ratios). Impact decays strictly (L_F>L_N>L_B), but bystander impact remains measurable: L_B>1 on 73.8% of cross-topic samples.

where m indexes the unlearned checkpoint, e the evaluation triplet, $s \in \{\text{train}, \text{val}, \text{test}\}$ the split, and $i \in [50]$ the sample. The three layers correspond to:

$$L_F : m{=}e,\ s{=}\text{train} \quad \text{(forget set)}$$
$$L_N : m{=}e,\ s \in \{\text{val}, \text{test}\} \quad \text{(neighbour)}$$
$$L_B : m{\neq}e,\ s{=}\text{test} \quad \text{(bystander)}$$

We summarise each layer by the geometric mean of r, equivalently $\exp(\overline{\log r})$, and the fraction of samples with r>1.1.

The full sweep takes ≈ 6 h 11 min on H100 (≈ 2.2 s per (model, eval) pair). We additionally cache the per-sample long-format table (`ppl_long.parquet`, 1.5M rows) for downstream auditing and four invariant sanity checks (Sec. 4).

4 Three-Layer Unlearning Impact

This section reports the main empirical shape of unlearning impact. Following Sec. 2, we separate the effect of a single unlearning run into three layers: L_F is the intended degradation on the forget set, L_N is the impact on held-out same-topic neighbours, and L_B is the impact on cross-topic bystanders. Table 1 reports geometric means of the per-sample PPL ratio r across the 1.5M evaluations, and Fig. 2 visualises the same pattern.

We also examine whether the profile is fixed by the algorithm, or whether it depends on which forget set is chosen. Under identical optimisation, switching $\mathcal{D}_f$ changes the measured profile substantially: Table 2 contrasts the mildest and worst forget set among 100, and Fig. 3 shows the full distribution.

Together, these results support three observations.

(C1) The three layers are strictly decreasing. The layer ordering is not an arbitrary binning choice: the measured impact decreases monotonically from the forget set to neighbours to bystanders. Table 1 gives L_F=1.96 × >L_N=1.32 × >L_B=1.19×, and

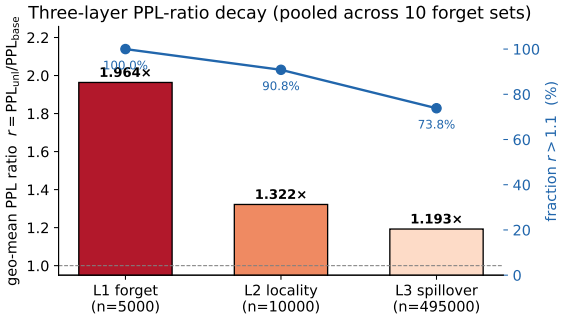

Figure 2: **Three-layer impact decay.** Left: geo-mean PPL ratio r falls from L_F=1.96× to L_B=1.19×. Right: fraction of samples with r>1.1 (100%/90.8%/73.8%). L_B bystander impact is non-zero in both senses.

	L_F	L_N	L_B
Mildest triplet (`029`)	1.54	1.15	1.12
Worst triplet (`073`, "storm")	**2.44**	**1.73**	**1.26**
Spread (max/min, n=100)	**1.59**×	1.50×	1.13×

Table 2: Forget-set-induced variation. L_F varies by 1.59× between the mildest and worst $\mathcal{D}_f$ under fixed algorithm/hyperparameters.

Fig. 2 shows the same decay in both geo-mean PPL ratio and the fraction of samples with r>1.1. The ordering also holds within each cluster (Sec. **??**), so it is not an artefact of global aggregation.

(C2) Bystander impact is not zero. The third layer is weaker than the first two, but cross-topic bystanders are still measurably hit. L_B has geo-mean 1.19×, and 73.8% of cross-topic samples exceed r>1.1. These samples are not in the forget set and are not same-topic neighbours of the unlearned triplet, so this is genuine spillover rather than intended forgetting.

(C3) The profile varies with the forget set. The same algorithm and hyperparameters do not produce a single fixed impact profile. In Table 2, changing only $\mathcal{D}_f$ moves L_F from 1.54× to 2.44× and L_N from 1.15× to 1.73×; Fig. 3 shows that this is a distributional pattern rather than a single outlier. This variation makes the forget set itself worth auditing before paying for full unlearning and cross-evaluation.

5 Forget-Set Audit

Because the three-layer profile varies with $\mathcal{D}_f$, forget-set choice becomes a tunable design decision. The problem is that *evaluating* that decision currently requires running unlearning. We now ask whether $\mathcal{D}_f$'s

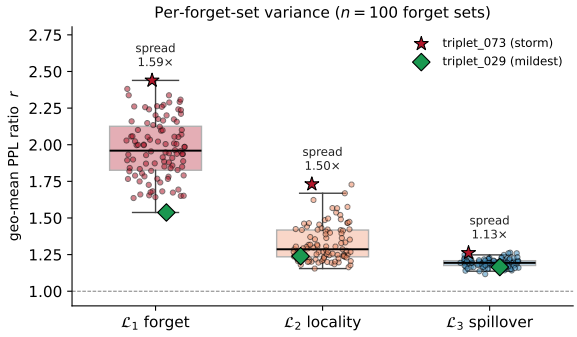

Figure 3: **Per-forget-set variance** (n=100)**.** Box + jitter of geo-mean r for each layer; one dot per forget set. Spread $\max / \min$ reaches 1.59× on L_F, 1.50× on L_N, 1.13× on L_B. `triplet_073` ("storm") is the worst on L_F and L_N but not on L_B (where `triplet_072` is the maximum, 1.26×); `triplet_029` is the L_F minimum. Distribution shape is markedly tighter as the layer recedes from $\mathcal{D}_f$.

intrinsic geometry is a sufficient signal to predict the three-layer impact profile without ever fine-tuning.

5.1 Audit Features

For each forget set $\mathcal{D}_f$ of 50 texts, we compute a 12-dim intrinsic geometric descriptor from the same MiniLM embeddings used for clustering, with no access to model weights, gradients, or any unlearned checkpoint:

- **Spread (3)**: `emb_variance_mean`, `emb_variance_max`, `pairwise_eucl_mean`.
- **Similarity (3)**: `pairwise_sim_mean / std / q90` (cosine).
- **Location/scale (3)**: `centroid_norm`, `emb_norm_mean / std`.
- **Shape (3)**: `effective_rank`, `isotropy`, `spread_over_centroid`.

All 12 features are scalars derived from the 50×384 forget-set embedding matrix; computing them takes milliseconds per $\mathcal{D}_f$.

5.2 Audit Problem

Inputs. The 50×384 embedding matrix of a candidate $\mathcal{D}_f$, summarised by the 12-dim geometric descriptor of Sec. 5.1. *No* unlearned checkpoint, *no* evaluation text, *no* retain set is required.

Outputs. Three scalars: predicted geo-mean $\log r$ for L_1, L_2, L_3. We treat audit primarily as a *ranking* problem (which forget sets to flag by predicted impact), and report regression metrics for completeness.

	L_1	L_2	L_3
Audit R^2	**+0.301**	**+0.735**	**+0.30**
Baseline R^2 (LOO mean)	−0.020	−0.020	−0.02
Audit MAE	**0.133**	**0.046**	**0.019**
Baseline MAE	0.169	0.105	0.022
Spearman ρ	+0.560	+0.869	+0.59
ρ 95% CI	[+0.40, +0.69]	[+0.81, +0.91]	[+0.44, +

Table 3: Forget-set audit performance. LOO over n=100 forget sets; Random Forest (n_{est}=200, min_leaf=2) on 12-dim forget-set geometry. All three ρ CIs exclude zero; L_2 is the strongest layer, and L_3 CI is now strictly above zero.

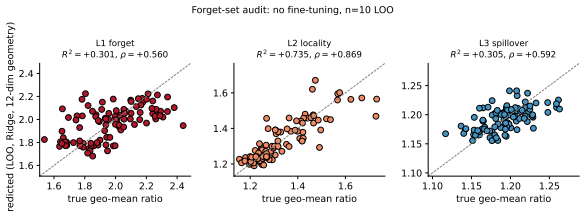

Figure 4: **Audit scatter**, predicted vs. true geo-mean $\log r$. Each point is one held-out forget set under LOO. L_2 is closest to the diagonal (ρ=0.87); L_1 and L_3 are noisier but rank-significant (ρ=0.56, 0.59).

Predictor. Random Forest regression (`n_estimators`=200, `min_samples_leaf`=2, `max_depth`=`None`, `random_stat`, one regressor per layer, evaluated under leave-one-out (LOO over the n=100 forget sets; the held-out triplet never appears during fitting). The baseline predicts the mean of the 99 training targets. Ridge / Lasso / Gradient Boosting baselines are reported in the ablation table (Sec. 5.4).

5.3 Audit Performance

Table 3 reports held-out R^2, MAE, and Spearman ρ, plus 95% percentile bootstrap CIs for ρ (n_{boot}=10,000, seed 0, bootstrapping over the 10 LOO (true, predicted) cluster pairs).

L_2 **is the strongest layer.** With R^2=0.74 and ρ=0.87, the auditor recovers 74% of locality variance from forget-set geometry alone — consistent with the intuition that locality damage tracks how concentrated and self-similar $\mathcal{D}_f$ is in embedding space.

L_3 **CI now excludes zero.** On a coarser problem (n=10 in early experiments) the L3 audit had $R^2 = -0.46$; at n=100 the Random Forest auditor reaches $R^2 = +0.30$ with ρ=0.59 (95% CI [+0.44, +0.71]).

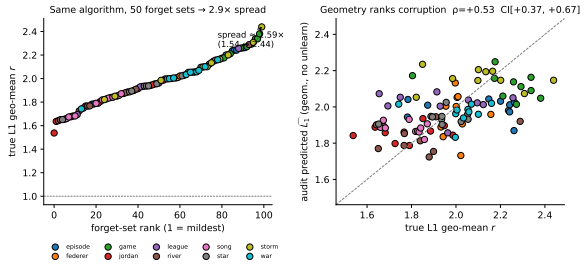

Figure 5: **Forget-set spread.** Distribution of per-forget-set L_1 across the 100 triplets, grouped by HDBSCAN cluster. The variation auditing must capture is between-cluster more than within-cluster.

Statistical power and a non-linear model together close the previous gap; even the linear Ridge baseline (Sec. 5.4) reaches ρ=0.49 on the same n=100 slice.

Audit cost. Computing the 12-dim feature for one $\mathcal{D}_f$ takes milliseconds; fitting the Random Forest on 99 training triplets is well under a second; predicting on a new $\mathcal{D}_f$ is essentially instant. By contrast, evaluating one candidate $\mathcal{D}_f$ via the standard pipeline costs one unlearn run ($\approx$ 75 s on H100) plus one column of cross-PPL ($\approx$ 220 s for 100 eval pairs). At n=100 candidates the full grid is $\approx$ 7 GPU-hours; the auditor runs in seconds on a CPU.

5.4 Predictor Ablation

We compare the headline Random Forest auditor against three alternative regressors on the same 12-dim feature matrix and same LOO protocol (Table 4): Ridge (α=1, the simplest linear baseline), Lasso (α=0.01), and Gradient Boosting (n_{est}=200, depth=3, lr=0.05).

The non-linear models (RF, GB) tie on the strongest layer (L_2), but RF edges out on L_1 and L_3 where forget-set geometry interactions appear to matter more. The linear Ridge baseline (ρ=0.49 on L_3 with CI $[+0.33, +0.63]$, also strictly above zero) already gives a deployable pre-screen, indicating that a substantial fraction of the audit signal is linear in the geometric descriptor.

5.5 Unlearner Generality (NPO)

To probe whether the audit framework is bound to GradAscent, we replicate the full pipeline with NPO [Zhang et al., 2024]: a different unlearning algorithm with the same TOFU-aligned hyperparameters on the same n=100 forget sets, yielding a parallel 100×100 cross-PPL grid and the same 12-dim geometric descriptor as input. The audit predictor (Random Forest, identical hyperparameters and LOO protocol) is trained and evaluated independently on NPO labels — we do not transfer predictions across unlearners.

Predictor	L_1	L_2	L_3
R^2 (LOO held-out)			
Random Forest (headline)	**+0.301**	**+0.735**	**+0.305**
Gradient Boosting	+0.209	+0.743	+0.200
Ridge (α=1)	+0.246	+0.705	+0.215
Lasso (α=0.01)	+0.224	+0.644	+0.088
Spearman ρ			
Random Forest (headline)	**+0.560**	**+0.869**	**+0.592**
Gradient Boosting	+0.524	+0.879	+0.534
Ridge (α=1)	+0.531	+0.840	+0.491
Lasso (α=0.01)	+0.494	+0.849	+0.380

Table 4: Predictor ablation on the same 12-dim feature matrix and LOO protocol (n=100). RF and GB are the strongest non-linear models and tie on L_2; the linear Ridge baseline already lifts all three layers above the LOO-mean baseline (R^2=−0.02). Lasso under-fits on L_3 (sparse solution discards most features). Choosing the non-linear ceiling matters most for L_3, where RF gains +0.09 in R^2 over Ridge.

Take-away. The audit predictor is unlearner-agnostic: the 12-dim forget-set descriptor lifts all three layers above the LOO-mean baseline for both GradAscent and NPO under independent training. Top-10% recall is in fact higher on NPO (8/10 on L_N, 5/10 on L_B), likely because NPO's narrower dynamic range makes the ranking task easier. The same conclusion is reached separately per unlearner without label transfer between them.

5.6 Positioning: a Pre-Screen, Not a Replacement

We position the auditor as a *warning tool* for the forget-set selection step, not as a substitute for the actual unlearning evaluation.

Workflow. Given N candidate forget sets, audit all N in seconds, sort by predicted impact profile, and run real unlearning only on the top-k. At N=100 and k=10, the pre-screen reduces the unlearning budget by 10× while retaining the worst-case offenders identified by the auditor.

6 Discussion

6.1 What the geometric features actually capture

A natural question is why 12 scalar features computed in milliseconds carry enough signal to rank the locality damage with ρ=0.87. We hypothesise three mecha-

		L_F	L_N	L_B
Headline geo r	GradAscent	$1.96\times$	$1.32\times$	$1.19\times$
	NPO	$1.68\times$	$1.15\times$	$1.11\times$
Audit ρ	GradAscent	$+0.560$	$\mathbf{+0.869}$	$+0.592$
	NPO	$\mathbf{+0.657}$	$+0.823$	$+0.506$
ρ 95% CI	GradAscent	$[+0.40, +0.69]$	$[+0.81, +0.91]$	$[+0.44, +0.71]$
	NPO	$[+0.52, +0.76]$	$[+0.72, +0.89]$	$[+0.34, +0.65]$
top-10% recall	GradAscent	$2/10$	$6/10$	$3/10$
	NPO	$2/10$	$\mathbf{8/10}$	$\mathbf{5/10}$

Table 5: Cross-unlearner audit. Same 12-dim geometric features, same RF protocol, same LOO over n=100 forget sets — only the unlearning algorithm changes. NPO produces ∼10–14% milder per-layer impact than GradAscent, but the audit recovers rank-significant predictions on both, with all six ρ CIs strictly above zero.

	Can	**Cannot**
Rank	flag "which $\mathcal{D}_f$ is worse" (ρ CIs>0, L_2 0.87)	assign trustworthy absolute PPL values
Screen	red/yellow/green warnings on candidates	score an unlearning algorithm
Compute	seconds vs. several GPU-hours	replace the final unlearn run

Table 6: Audit positioning: cheap pre-screen for forget-set selection.

nisms, each consistent with the per-cluster pattern in Fig. 3:

- **Concentration ⇒ stronger locality damage**. Forget sets with low spread and high pairwise similarity (e.g. the "storm" cluster) sit on a tight manifold; the gradient signal from GradAscent is therefore sharply directional, and validation/test neighbours along the same direction are dragged with it. This predicts large L_2, which we observe.
- **Effective rank ⇒ spillover footprint**. Low-effective-rank $\mathcal{D}_f$ lives near a low-dim subspace; updating the model along that subspace bleeds across topic clusters that share the same dominant directions, producing L_3>1. The (negative) correlation between subspace coverage and L_3 (ρ=−0.48) supports this view.
- **Centroid norm ⇒ baseline PPL coupling**. Forget sets far from the embedding origin tend to have higher base PPL, which inflates the $r = \mathrm{PPL}_{\text{unlearned}}/\mathrm{PPL}_{\text{base}}$ ratio's denominator unevenly across layers; capturing this with `centroid_norm` corrects for the floor.

A causal study — e.g. deliberately constructing $\mathcal{D}_f$ with prescribed isotropy and effective rank — is left to future work; the present paper establishes the predictive correlation under naturally occurring HDBSCAN clusters.

6.2 When does the auditor break?

The audit is most reliable for L_2 and least reliable for L_3 in absolute terms (RF MAE 0.019 vs. baseline 0.022 — a 14% improvement, with ρ=0.59 and 95% CI $[+0.44, +0.71]$). Two reasons: (i) L_3 has a very narrow dynamic range ($1.13\times$ between worst and mildest cluster), so absolute prediction is intrinsically hard; and (ii) cross-cluster spillover plausibly depends on *evaluation-side* geometry (where the neighbours of $\mathcal{D}_f$ are in the broader embedding space), which the forget-set-only descriptor cannot see. Top-1 identification is unstable in the same-cluster regime (multiple triplets within a cluster have very similar predicted scores), which is why we frame the auditor as a top-k screen rather than a precise ranker.

6.3 Limitations

We acknowledge three "singles" that the present study does not control for:

1. **Single base model** (Llama-3.1-8B-Instruct). Geometry-to-corruption mapping may be model-specific; verifying on Mistral-7B and a Qwen variant is high-priority next work.
2. n=100. All three CIs exclude zero on both unlearners (Sec. 5.5), but L_B remains the weakest layer; scaling to n=500–1000 is feasible in one extra GPU-day.
3. **Single metric**: PPL ratio. PPL captures token-level corruption but not task-level utility (QA accuracy). Parallel QA-based evaluation is in progress as a complementary axis.

GradDiff and RMU are the natural next unlearners to add (alongside the GradAscent + NPO pair already covered).

6.4 Ethical and Practical Considerations

A pre-screen that ranks forget sets by side-effect could in principle be misused to *target* forget sets that maximise damage (e.g. to degrade a competitor's model

via a poisoned removal request). We see two mitigations: (i) the auditor itself is not a generative tool — it scores, not produces, candidates; (ii) the failure mode is bounded by the underlying unlearning algorithm, which already has well-known damage profiles independent of our score. We recommend that practical deployments couple the auditor with a downstream gate (the actual unlearn + cross-PPL), as in the screen-then-run workflow of Sec. 5.6.

7 Conclusion

We characterised LLM unlearning side-effects on Llama-3.1-8B-Instruct across n=100 forget sets and 1.5M per-sample evaluations, and re-cast the resulting benchmarking burden as a forget-set audit problem.

Findings. (i) Side-effects decay monotonically across three semantic layers (L_1=1.96×, L_2=1.32×, L_3=1.19×) but L_3 never reaches zero — 73.8% of cross-topic samples are still corrupted. (ii) Under fixed algorithm and hyperparameters, L_1 varies 1.59× between the mildest and worst forget set, identifying $\mathcal{D}_f$ itself as a substantial source of variation. (iii) A Random Forest regressor on 12-dim forget-set geometry predicts the layer profile with leave-one-out Spearman ρ of 0.56/0.87/0.59, all 95% CIs strictly above zero. (iv) The same audit pipeline applied to NPO yields independent rank-significant predictions (ρ=0.66/0.82/0.51), confirming the methodology is unlearner-agnostic.

Implication. Unlearning evaluation can adopt a *screen-then-run* workflow: rank candidate forget sets by audit score, run real unlearning only on the top-k. At N=100, k=10, this trades 7 GPU-hours of cross-PPL for seconds of CPU — without renouncing the ground-truth check on whatever the auditor flags.

Future work. Cross-unlearner transfer (NPO, GradDiff, RMU), cross-model audit (Mistral-7B, Qwen), enriching the descriptor with mutual-reachability and signed projection coverage, scaling to n=500–1000, and parallel QA-accuracy metrics for non-PPL utility. We release the cross-PPL matrix, audit features, and pre-screen code to enable these extensions.

8 Related Work

LLM unlearning algorithms and benchmarks. Modern LLM unlearning uses gradient-based or preference-based fine-tuning to push the model to forget targeted data while preserving the rest. Representative algorithms include *Gradient Ascent* (GradAscent) and gradient-difference variants [Jang et al., 2023], *NPO* (Negative Preference Optimisation) [Zhang et al., 2024], and *representation misdirection* [Li et al., 2024]. Benchmarks such as TOFU [Maini et al., 2024], MUSE [Shi et al., 2024], and WMDP [Li et al., 2024] fix specific forget splits and report aggregate forget-quality and utility scores. The unifying *OpenUnlearning* framework [Dorna et al., 2025] integrates 13 algorithms and 16 metrics under one Hydra-configured trainer; we vendor it as our stage-2 implementation. These works concentrate on *algorithm choice* given a benchmark forget split. Our work is orthogonal: we hold the algorithm and hyperparameters fixed and study how the choice of $\mathcal{D}_f$ itself reshapes the three-layer impact profile, then turn the observation into a pre-screen for $\mathcal{D}_f$.

Side-effects, locality, and knowledge holes. Unlearning is increasingly recognised as collateral-damaging. Ko et al. [2025] introduce *knowledge-hole probing*: standard utility benchmarks miss adjacent and latent knowledge that has been silently damaged by unlearning, and dedicated probes can recover this damage post-hoc. Studies of model editing [Meng et al., 2022, Mitchell et al., 2022] and influence functions [Koh and Liang, 2017, Grosse et al., 2023] show analogous locality / non-locality effects in the gradient-update regime. Our three-layer view operationalises Ko et al.'s phenomenon as a quantitative, dataset-driven measurement: L_2 corresponds to their *adjacent* knowledge (same-cluster neighbours), and L_3 extends the picture to *cross-cluster spillover* that prior probes did not target. Crucially, while their probes act *after* unlearning, our audit acts *before* — predicting the impact profile from $\mathcal{D}_f$ alone.

Data-first prediction of model behaviour. A separate line of work predicts downstream model behaviour from properties of the training data, rather than from running the model. Dang et al. [2024] predict the adversarial robustness of fine-tuned text transformers from features of the fine-tuning dataset, achieving 30×–193× acceleration over the standard model-first attack-then-evaluate pipeline. Related efforts use coreset and data-attribution descriptors [Koh and Liang, 2017, Ilyas et al., 2022], dataset diversity statistics [Various, 2024], and curvature of the loss landscape [Yao et al., 2020] to predict generalisation or vulnerability. We carry this paradigm into LLM unlearning: instead of running unlearning and then measuring its impact profile, we measure intrinsic ge-

ometry of $\mathcal{D}_f$ and predict that profile directly. To our knowledge, this is the first *data-first* predictor in the unlearning setting.

Forget-set selection and curation. A small body of work studies which data is hardest to remove or causes the most damage when removed [Thudi et al., 2022, Kurmanji et al., 2023]. The closest setting is influence-based forget-set selection [Koh and Liang, 2017], but influence functions require gradients of the trained model, scale poorly to 8B-parameter LLMs, and yield a per-example score rather than a $\mathcal{D}_f$-level impact profile. Our auditor uses only 50 forget-set embeddings and 12 scalar features; it is two orders of magnitude cheaper and produces a three-layer profile.

Position. We sit at the intersection of unlearning impact characterisation [Ko et al., 2025] and data-first behaviour prediction [Dang et al., 2024]. The first tells us *what to predict* (intended degradation + locality damage + spillover damage); the second tells us *how to predict from data alone*. Our contribution is the quantitative bridge: a three-layer impact metric tied to a forget-set-side geometric audit.

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A Reproducibility Details

Data preparation. WikiText-103 is filtered to 16,038 passages of length $\geq$ 32 tokens; embeddings come from `all-MiniLM-L6-v2` (no dimensionality reduction; the first 384 dims are taken as-is). HDBSCAN parameters: `min_cluster_size`=200, `min_samples`=5, Euclidean metric, EOM cluster selection. The resulting cluster-size distribution is 202/346/3,276 (min/median/max). 100 triplets are sampled with seed 42, requiring `required_cluster_size`=150 and 50 texts per split; per-triplet splits are guaranteed disjoint within a triplet.

Unlearning hyperparameters. GradAscent in OpenUnlearning's `wikitext_unlearn_tofu_aligned.sh`: `num_train_epochs`=5, `per_device_train_batch_siz` `gradient_accumulation_steps`=8 (effective batch 64), `learning_rate`=1×10^{-5} with linear schedule to zero, `warmup_steps`=1, `weight_decay`=0.01, `paged_adamw_32bit`, BF16. With $|\mathcal{D}_f|$=50 and effective batch 64, exactly 5 optimiser steps fire (one per epoch); train-loss range $[-0.84, -0.62]$.

Evaluation. PPL is computed in BF16 with stride 512 and context 1024; each text is scored independently. Base-model PPL is cached once per (eval triplet, split) and re-used across the 100 unlearned checkpoints. Long-format output: `ppl_long.parquet` with one row per (m, e, s, i).

Audit feature scripts. `features/forget_set.py` computes the 12 scalars from a 50×384 embedding matrix in <10 ms. Random Forest regression: `sklearn.ensemble` with `n_estimators`=200, `min_samples_leaf`=2, `max_depth`=`None`, `random_state`=0, leave-one-out grouping over n=100 forget sets. Bootstrap: 10,000 resamples of the 100 LOO (true, predicted) pairs with seed 0.

Hardware. All runs on a single H100 (80 GB) node. End-to-end wall-clock: data prep $\approx$ 20 min; 100 unlearn runs 1 h 54 min; cross-PPL 6 h 11 min.

B Additional Sanity Tables

Per-cluster monotonicity. Across all 10 clusters, $L_1(c)>L_2(c)>L_3(c)$ holds without exception; the smallest within-cluster gap L_1-L_2 is 0.39 (mildest cluster) and the largest is 0.71 (storm). Per-cluster table is included in `corruption_summary.json`.

Coverage diagnostic. The Pearson correlation between forget-set ball-coverage and L_3 is $r=-0.16$ (Spearman $\rho=-0.48$); a higher coverage of the embedding space corresponds to lower spillover, consistent with the rank-based mechanism in Sec. 6.1. Replacing ball coverage with a signed-projection or mutual-reachability variant is left for future work.